Moving Beyond p-Values: A Bayesian Approach to GWAS

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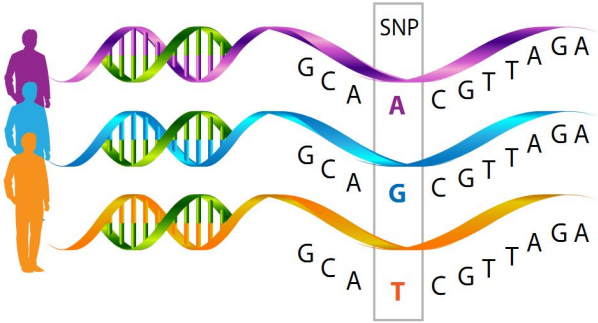
What is a Genome-Wide Association Study (GWAS)?

- Definition
- Necessary vocabulary
  - Allele / minor allele
  - Single nucleotide polymorphism (SNP)



Motivation

- Build off a GWAS framework that we learned in *STAT 494*
- Implement a new GWAS method that allows for easily interpreted results

Bayesian statistical methods for genetic association studies

Matthew Stephens and David J. Balding*[†,§]

Frequentist approach to GWAS

- More variables (SNP's) than observations in our data ($P >> n$)
- Marginal regression analysis
  - $Y = B_0 + B_1 \times SNP_i$
- Multiple testing → risk of Type I Error
- P-values are difficult to interpret

Vision

Using prior knowledge and observed data, learn to **apply Bayesian regression models in Genome-wide association studies** to uncover more intuitive and informative conclusions than traditional frequentist approaches.

What is Bayesian Statistics? Part 1

Update our existing beliefs with new observations

What is Bayesian Statistics? Part 2

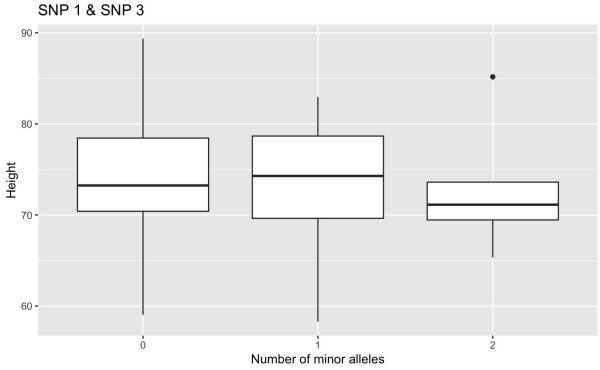
$$\underbrace{P(\theta \mid \text{data})}_{\text{Posterior}} = \frac{\overbrace{P(\text{data} \mid \theta)}^{\text{Likelihood}}\,\overbrace{P(\theta)}^{\text{Prior}}}{\underbrace{P(\text{data})}_{\text{Normalizing constant}}}$$

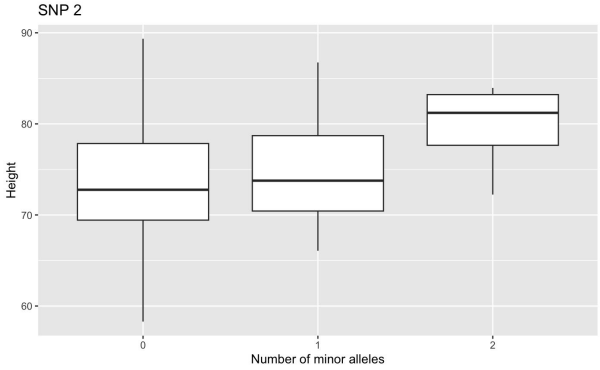
Implementation

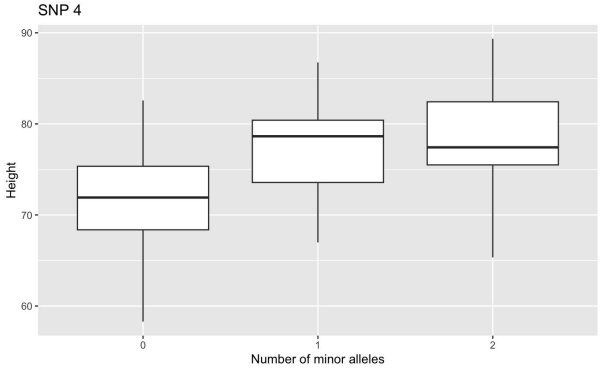
Data

Height	SNP 1	SNP 2	SNP 3	SNP 4
71	1	2	0	1
65	0	1	0	1
72	1	0	2	2

Visualizing the data







Methods

Model

$$Height_i = \theta + SNP_{Effect}\, x_i + \epsilon_i \quad \epsilon_i \sim N(0, \sigma^2)$$

Model

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$$\Big\downarrow \text{ Likelihood}$$

$$Height_i \mid \theta,\ SNP_{Effect},\ \sigma^2 \sim\ N(\theta + SNP_{Effect}\, x_i,\ \sigma^2)$$

Model

$$Height_i = \theta + SNP_{Effect}\, x_i + \epsilon_i \quad \epsilon_i \sim N(0, \sigma^2)$$

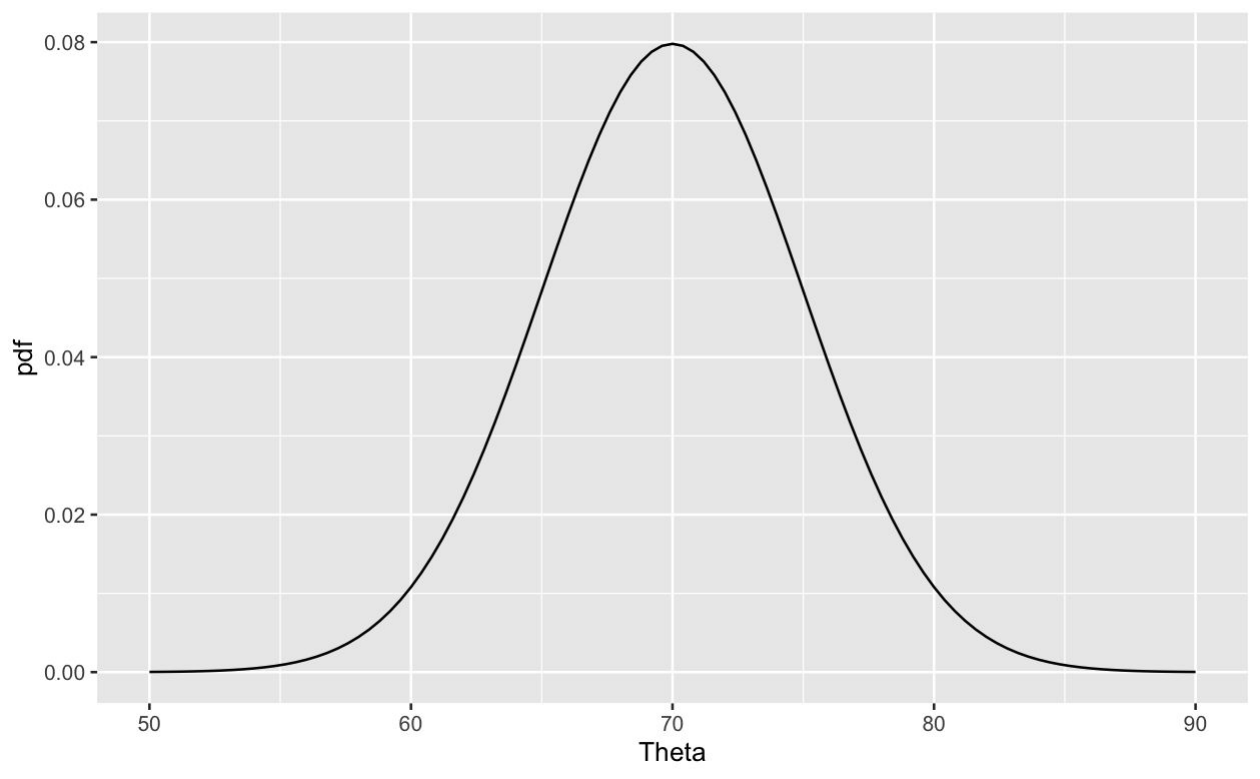
Likelihood

$$Height_i \mid \theta,\ SNP_{Effect},\ \sigma^2 \sim\ N(\theta + SNP_{Effect}\, x_i,\ \sigma^2)$$

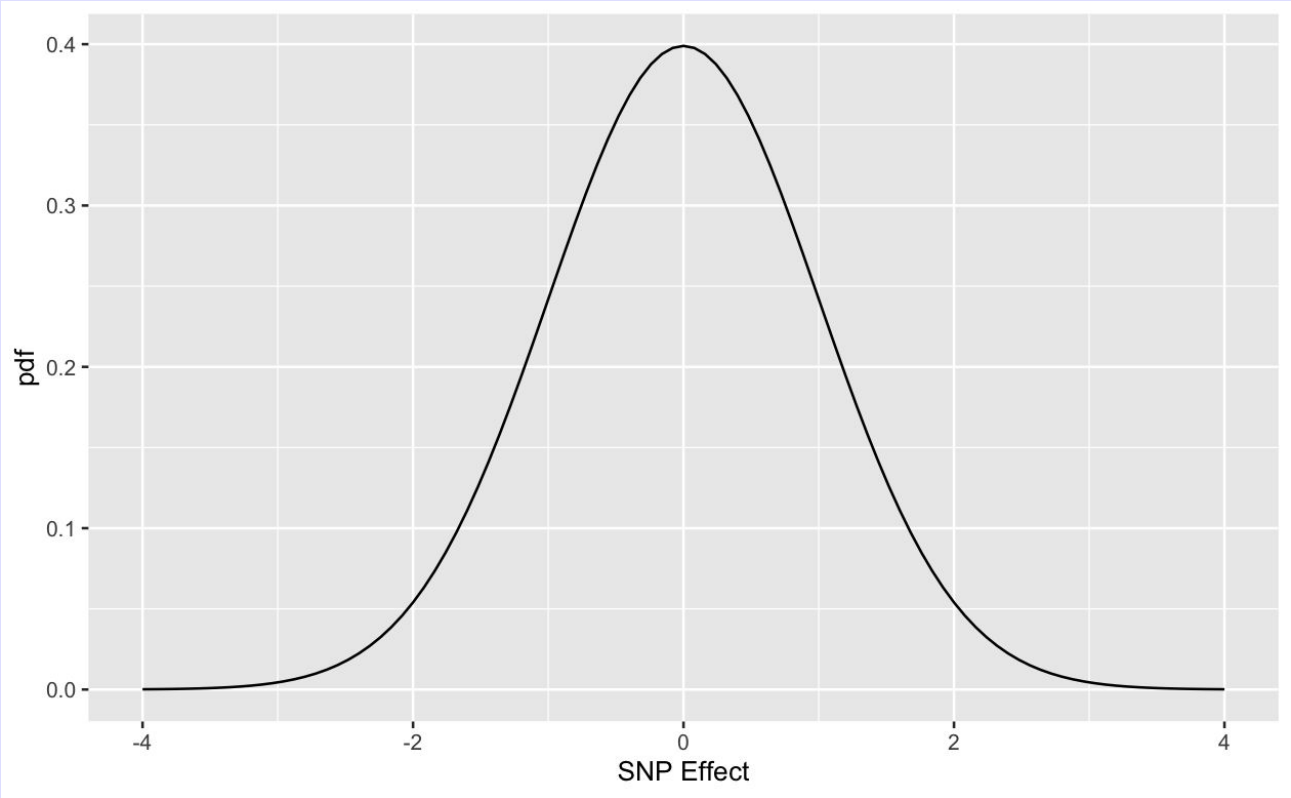
Priors

$$\theta \sim\ N(70, 5^2),\ SNP_{Effect} \sim\ N(0, 1),\ \sigma^2 \sim\ Exponential(0.2)$$

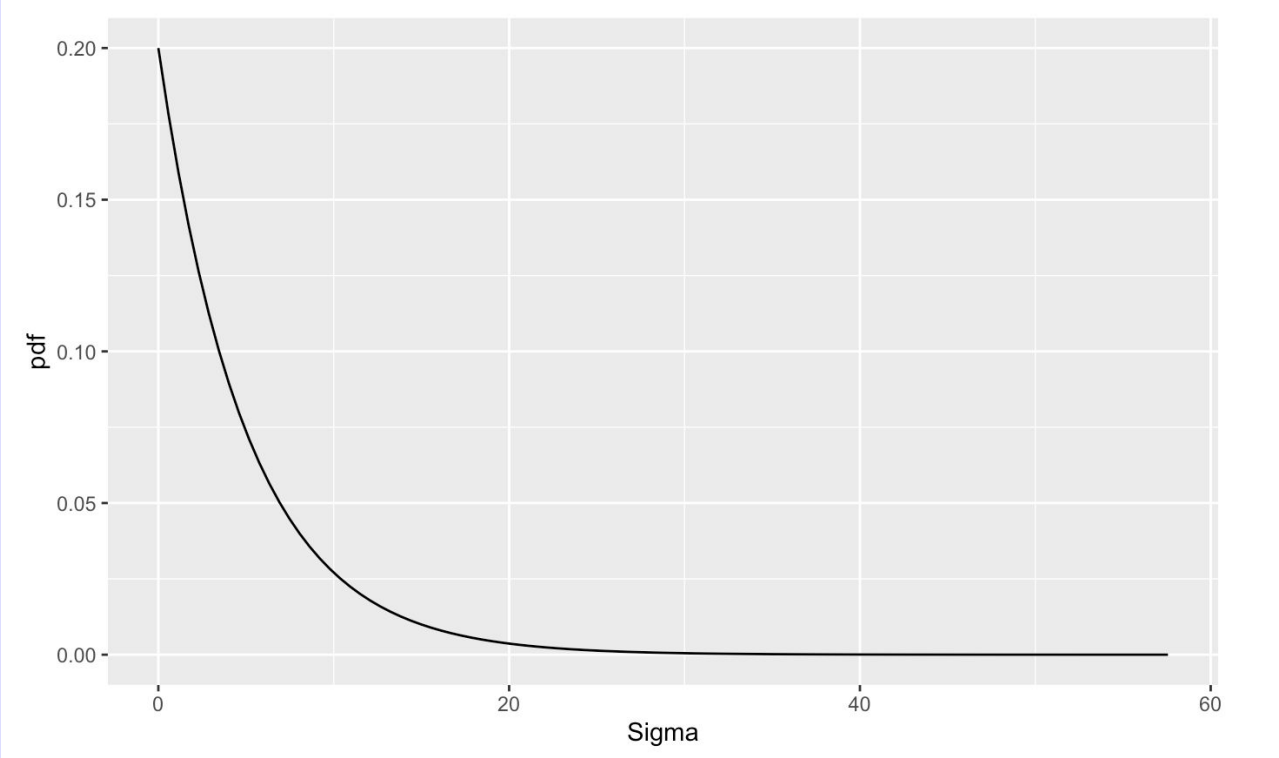
Prior - Intercept



Prior - Slope



Prior - Sigma squared



Putting it all together

$$\underbrace{\pi(\theta,\ SNP_{Effect},\ \sigma^2 \mid Height)}_{\text{Posterior}} \propto \underbrace{f(Height \mid \theta,\ SNP_{Effect},\ \sigma^2)}_{\text{Likelihood}}\ \underbrace{\pi(\theta)\ \pi(SNP_{Effect})\ \pi(\sigma^2)}_{\text{Priors}}$$

Normalizing constant

$$P(\theta \mid \text{data}) = \frac{P(\text{data} \mid \theta)\, P(\theta)}{\boxed{P(\text{data})}}$$

$$\boxed{\int \int \int f(Y \mid \beta_0, \beta_1, \sigma^2)\ \pi(\beta_0)\pi(\beta_1)\pi(\sigma^2) d\beta_0 \beta_1 d\sigma^2}$$

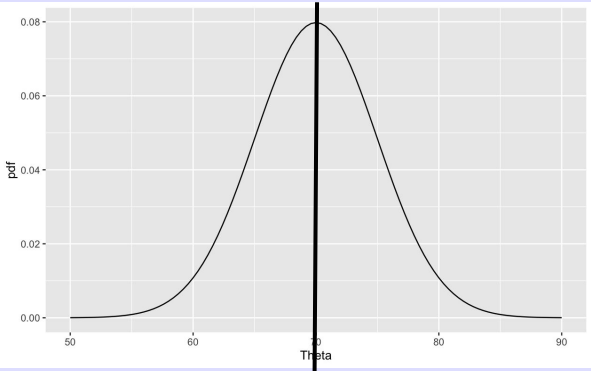
Posterior estimation

Markov Chain Monte Carlo (MCMC) → Hamiltonian Monte Carlo (HMC)

- $\pi(\theta,\ SNP_{Effect},\ \epsilon \mid Height)$ has no closed form solution
- **Core idea:**
  - Propose parameter values
  - Compute posterior plausibility
  - Move towards regions of higher plausibility

Results/Posteriors

Reminder of priors



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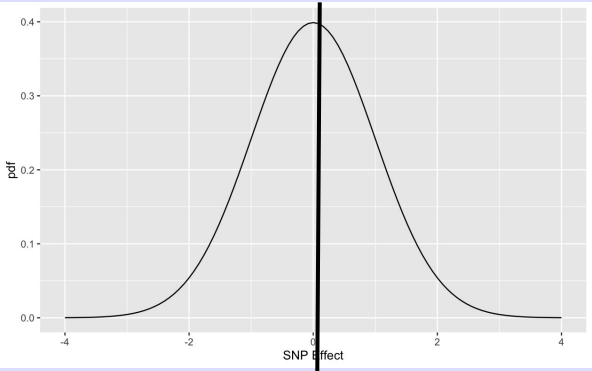

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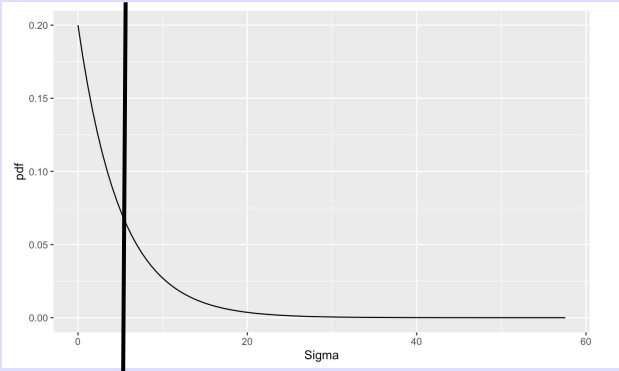

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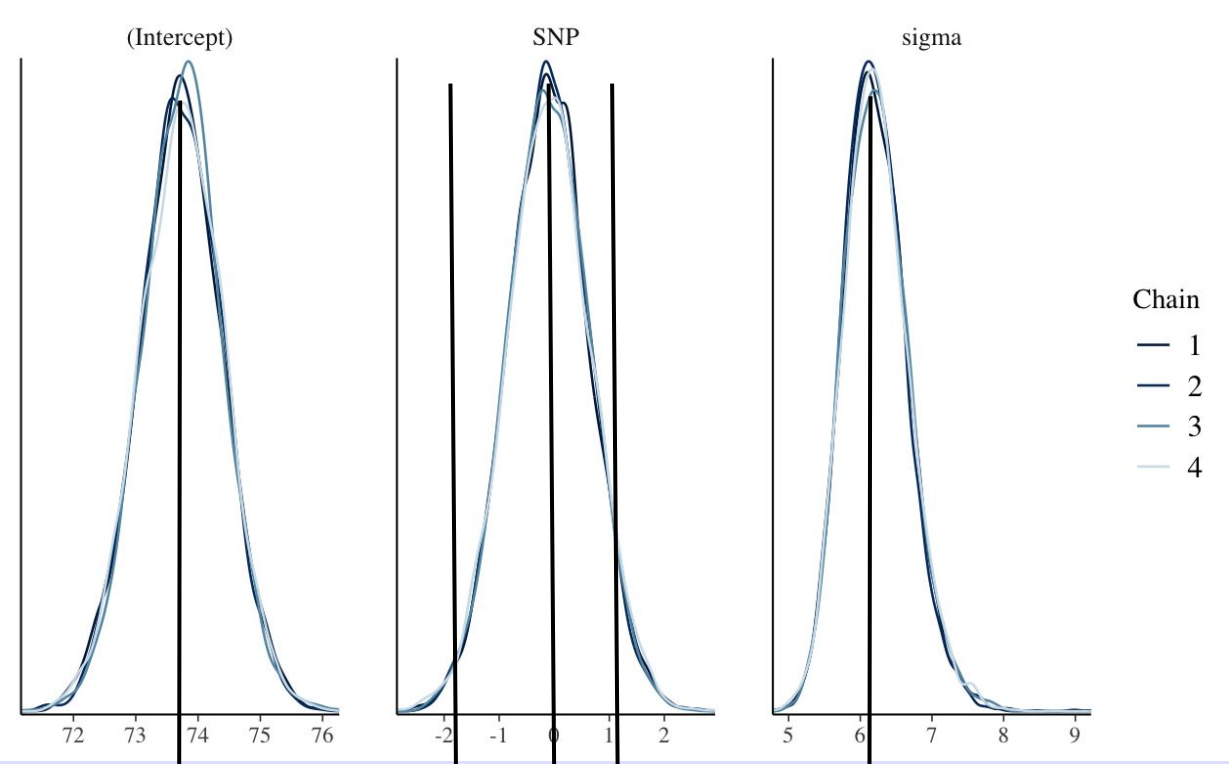
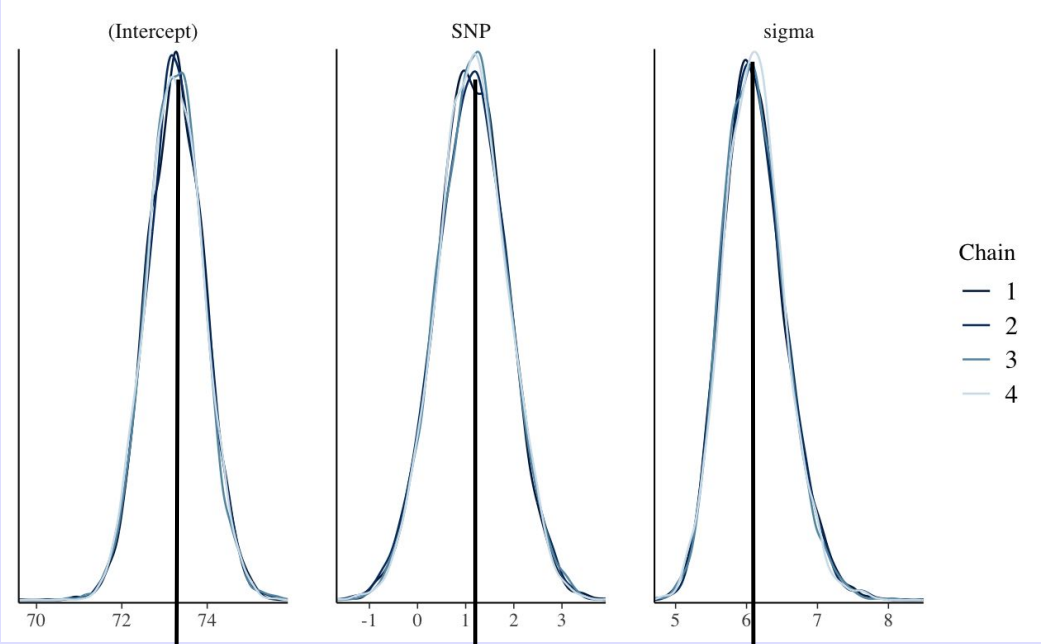
The only thing that changes now from SNP to SNP is our likelihood function (our observed data)

Results #1



~74 0 ~6

Results #2



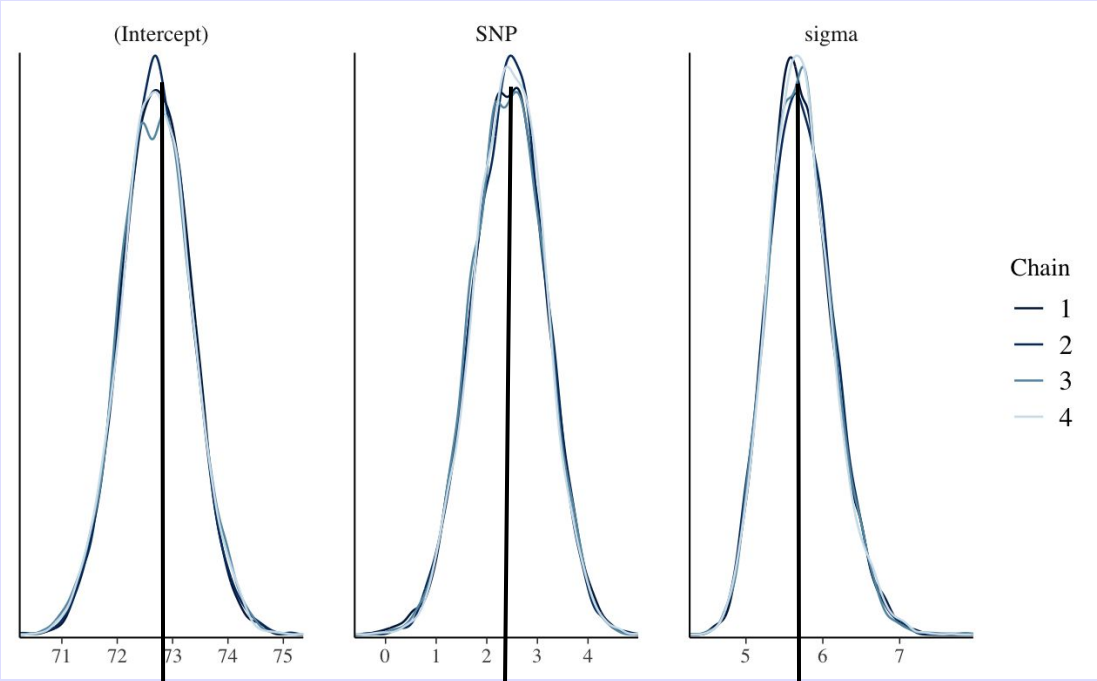
~74

~1

(-0.3 , 2.59)

~6

Results #3



~74

2.5
(1.03 , 3.8)

~6

Broad results

- Direct probabilistic statements
- Intuitive "updating" of knowledge

Limitations

- Computationally expensive
- Intellectually expensive
- Subjectivity

Sources

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Thank you!